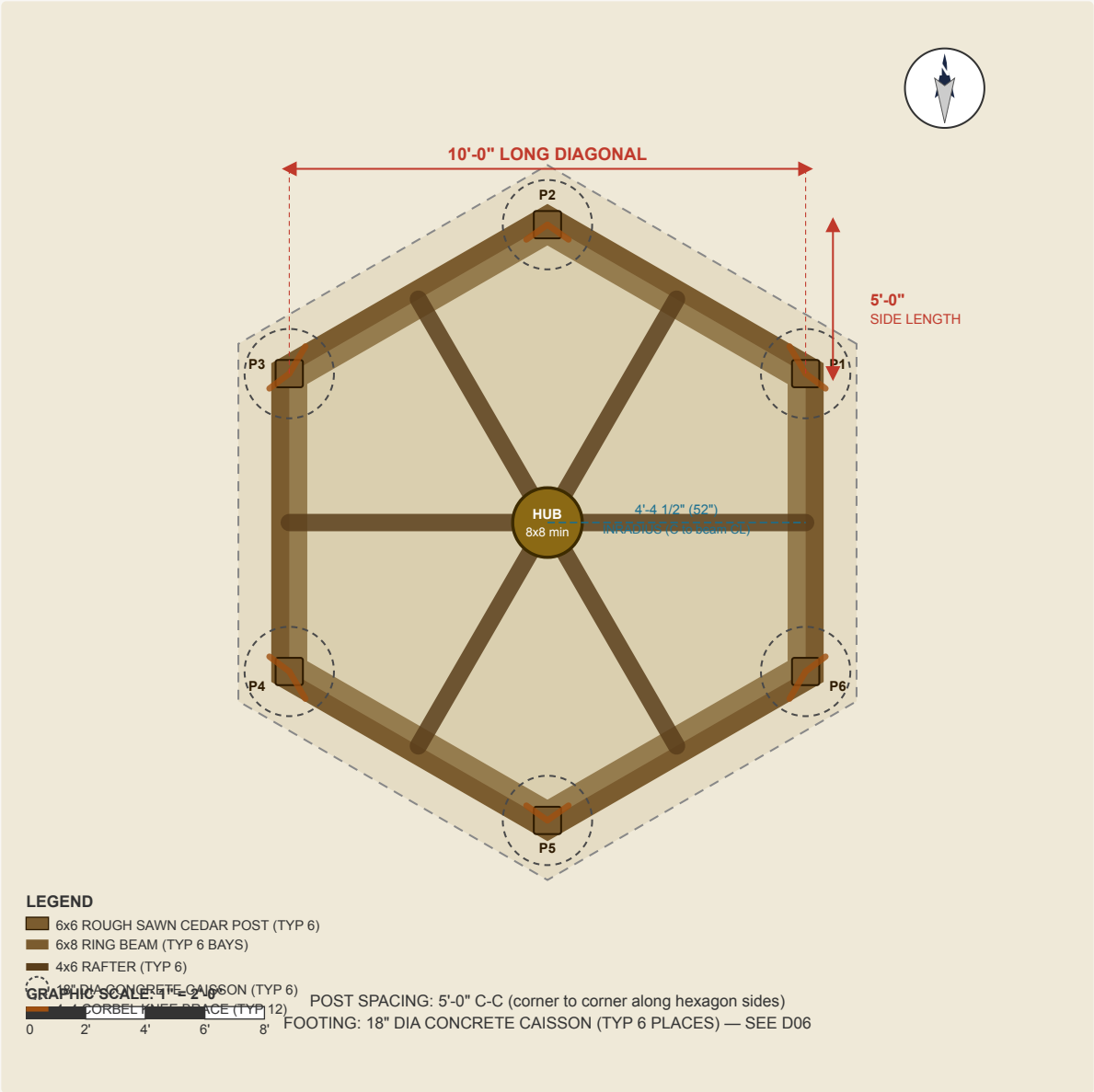
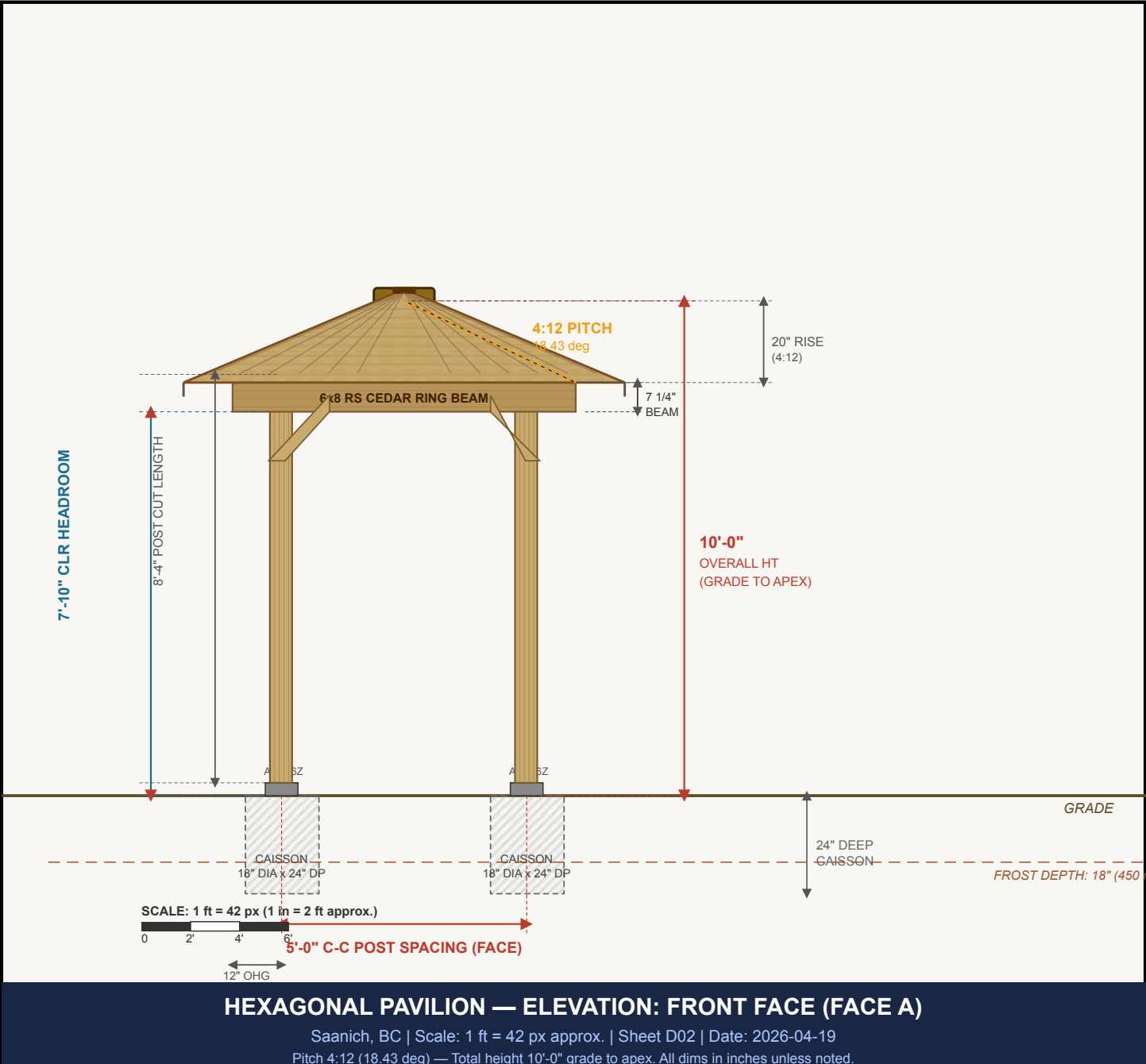
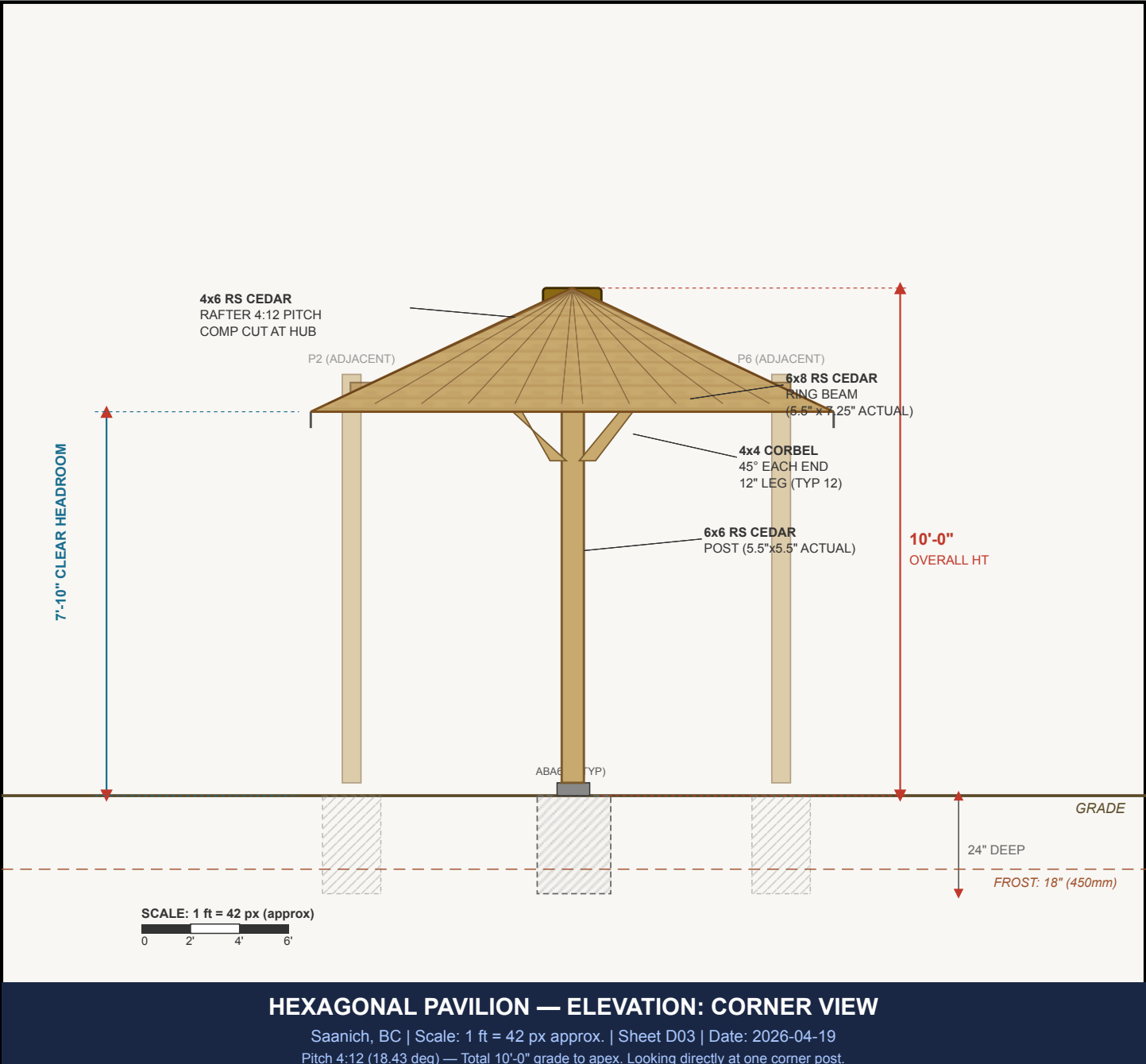




HEXAGONAL PAVILION — PLAN VIEW (LOOKING DOWN)

Saanich, BC | Scale: 1:24 (1 in = 2 ft) | Sheet D01 | Date: 2026-04-19
All dimensions in inches unless noted. Rough sawn cedar throughout.

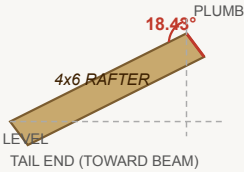




DETAIL A — RAFTER CUT GEOMETRY

1. PLUMB CUT AT TOP OF RAFTER (ELEVATION VIEW)

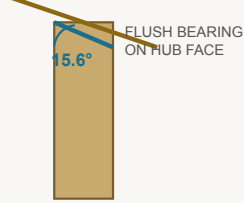
Miter angle = 18.43° (matches 4:12 roof pitch)



2. SIDE BEVEL AT TOP CUT (PLAN VIEW LOOKING DOWN)

Blade bevel = 15.6° — rafter cheek to fit flush on 60° hub face

HUB FACE (60° HEXAGON)



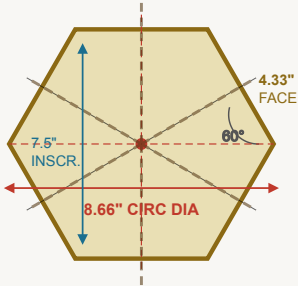
SAW SETUP — COMPOUND MITER SAW

Miter angle: 18.43° (4:12 pitch)
Blade bevel: 15.6° (hex face fit)
Feed direction: mark CROWN side up
All 6 rafter top cuts: IDENTICAL — do not reverse bevel
Test fit on scrap 4x6 before cutting all 6 pieces.
SDS screws: 2 per rafter, driven into hub after dry fit confirmed.

DETAIL B — HUB GEOMETRY

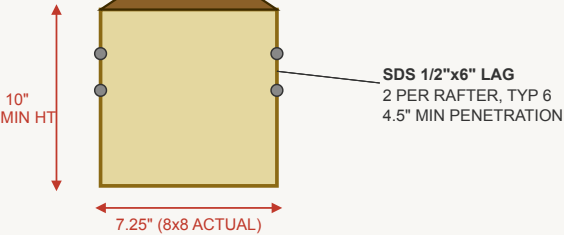
HUB — PLAN VIEW (LOOKING DOWN)

Regular hexagonal prism, 8x8 base stock (7.25" actual)
Six faces milled/planed to receive compound-cut rafter cheeks



HUB — SECTION / ELEVATION VIEW

Min 10 in tall. Lag screws driven from rafter into hub.



FABRICATION OPTIONS

Option A: Source 8x8 cedar turning blank, mill hex faces at millwork shop.
Option B: Laminate 3x 2x8 RS cedar. Glue + 3" GRK screws @ 6" o.c.
Hand-plane or saw 6 flat faces after glue-up cures (48 hr min).
Option C: Steel weld plate spider hub (fabricator quote separately).

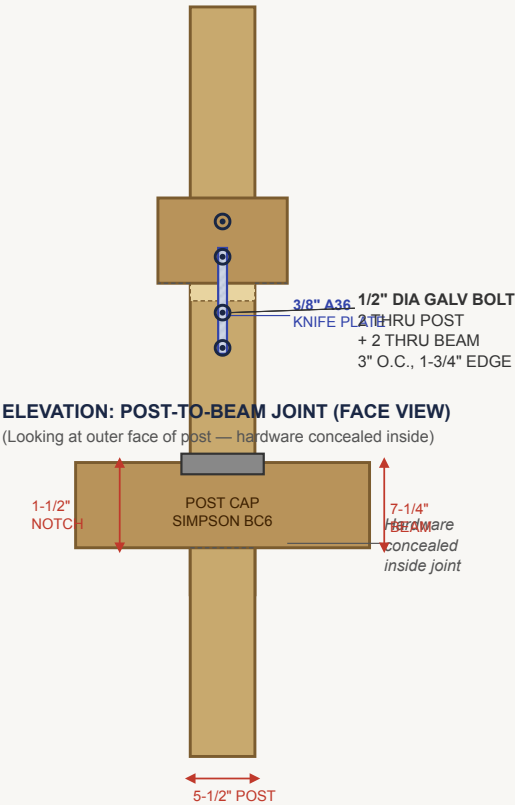
RAFTER COMPOUND CUT DETAIL + HUB GEOMETRY DETAIL

Saanich, BC | Scale: FULL SIZE (1:1) / 1:4 hub | Sheet D04 | Date: 2026-04-19

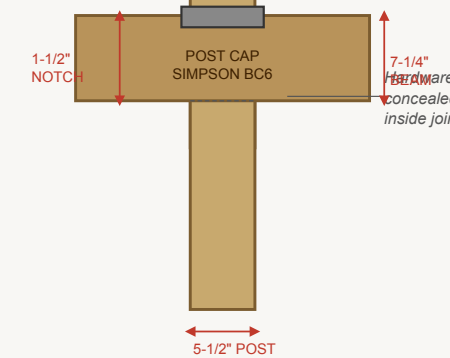
CRITICAL: Set compound miter saw at 18.43° miter + 15.6° bevel. Test on scrap first. All 6 top cuts identical.

DETAIL C — POST-TO-BEAM KNIFE PLATE

SECTION: KNIFE PLATE IN POST + BEAM
(Section cut through centre of post, looking along beam)

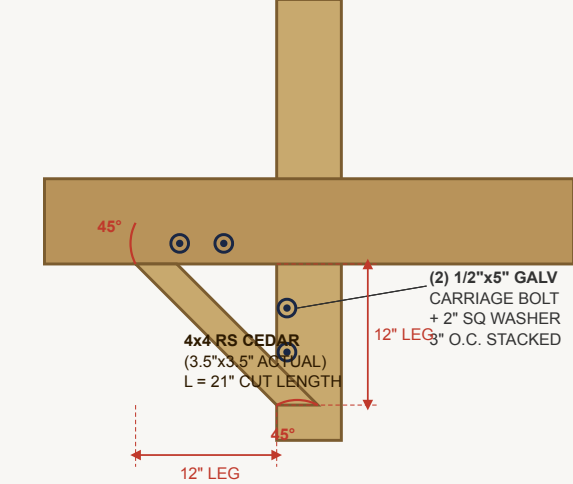


ELEVATION: POST-TO-BEAM JOINT (FACE VIEW)
(Looking at outer face of post — hardware concealed inside)

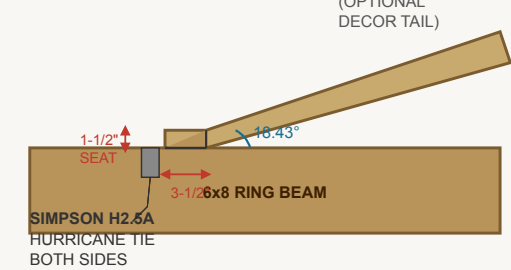


DETAIL D — CORBEL + BIRD'S MOUTH

CORBEL KNEE BRACE — ELEVATION
4x4 cedar, 45° each end, 12" leg each direction



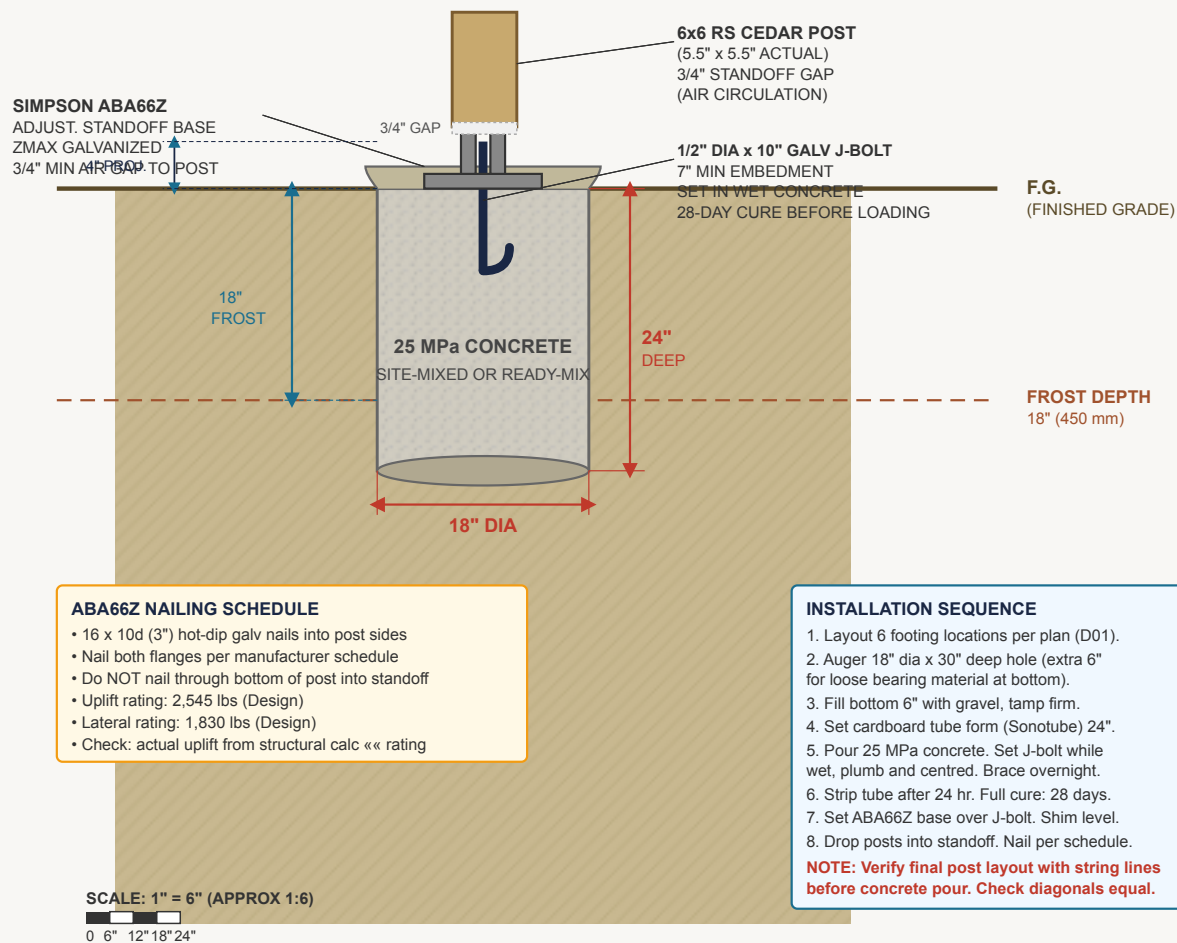
BIRD'S MOUTH RAFTER SEAT (AT RING BEAM)
Rafter 4x6, seat depth 1-1/2", plumb cut 18.43° AT TAIL (OPTIONAL DECOR TAIL)



SCALE: 1" = 8 px (APPROX 1:1.5)
0 4" 8" 12"

JOINERY DETAILS — POST-TO-BEAM / CORBEL / BIRD'S MOUTH

Saanich, BC | Scale: 1:4 (approx) | Sheet D05 | Date: 2026-04-19
Knife plate: 3/8" A36 steel. All bolts: 1/2" dia galvanized. Edge dist: 1-3/4" min.



FOOTING DETAIL — 18" DIA CONCRETE CAISSON + ABA66Z POST BASE

Saanich, BC | Scale: 1:6 (approx) | Sheet D06 | Date: 2026-04-19

25 MPa concrete. Frost depth 450 mm. 610 mm (24") embedment. TYP 6 PLACES.

SHOP BLUEPRINT — CUT LIST & ANGLE REFERENCE

Hexagonal Pavilion | Saanich, BC | BCBC 2018 | Date: 2026-04-19
All dimensions in INCHES unless noted. Rough sawn cedar throughout. Verify all dimensions before cutting.

ID	MEMBER	NOMINAL	ACTUAL (in)	QTY	CUT LENGTH	TOP CUT ANGLE	BOTTOM CUT	BF / TOTAL
CL-01	POST 6x6 RS Cedar	6x6	5.5 x 5.5	6	8'-4" (100")	Square (90°) Notch top: 1.5" dp x 7.25" w	Square (90°)	25.0 bf ea 150.0 bf
CL-02	RING BEAM 6x8 RS Cedar	6x8	5.5 x 7.25	6	5'-0" (60")	30° MITER Both ends same direction (set saw at 30° to CL)	30° MITER Hex exterior angle = 60° Miter saw at 30°	24.0 bf ea 144.0 bf
CL-03	RAFTER 4x6 RS Cedar	4x6	3.5 x 5.5	6	6'-4" (76") Order 7'-0" stock	COMPOUND CUT Miter: 28.7° Bevel: 9.1° Crown UP. All 6 identical. TEST CUT ON SCRAP FIRST	BIRD'S MOUTH Seat: 1.5" dp x 3.5" w Plumb cut: 18.43° + Tail plumb cut at 18.43° 12" overhang beyond CL	12.7 bf ea 76.0 bf
CL-04	HUB BLOCK 8x8 RS Cedar (or 3-ply 2x8)	8x8	7.25 x 7.25	1	12" rough block Mill hex faces on site	6 flat faces @ 60° Face width: 4.33" ea See D04 for geometry	Square bottom Finial or rain cap on top	8.0 bf 8.0 bf
CL-05	CORBEL 4x4 RS Cedar	4x4	3.5 x 3.5	12	21" (1'-9") Order 2'-0" stock	45° plumb cut No side bevel required	45° plumb cut Both ends same angle	2.33 bf ea 28.0 bf
CL-06	ROOF DECKING 1x6 T&G RS Cedar	1x6 T&G	0.75 x 5.5	By BF	Random lengths 6-10 ft, taper rip on site	Taper rip each plank 6 pie-wedge bays	Butt to hub, nail at each rafter crossing	105 bf w/ waste 105.0 bf
CL-07	CEDAR SHINGLES #1 Blue Label, 18"	#1 18" WRC	—	10 bdl	5.5" exposure	Apply per CSSB guidelines Stagger joints 1.5" min	Hip/ridge caps: 2 bdl	12 bdl total
TOTALS					Order long — trim to fit		STRUCTURAL FRAMING:	476.5 bf nominal
Add 15% waste to all BF totals when placing lumber order.							WITH 15% WASTE:	~549 bf to order

ANGLE REFERENCE QUICK CARD — FOR SAW SETUP

RAFTER — COMPOUND CUT (HUB END) Tool: Compound Miter Saw Miter Angle: 28.7° Blade Bevel: 9.1° Crown direction: UP (mark before cutting) Do NOT flip bevel — all 6 cuts identical Rafter tail (beam end): plumb cut only at 18.43° Bird's mouth seat depth: 1-1/2" (do not exceed) *** Test on scrap 4x6 before cutting production *** SDS lags: 2 per rafter, pre-drill 1/4" pilot into hub Rafter-to-beam: Simpson H2.5A hurricane tie + bird's mouth Horizontal run: 72" total (60" to hub + 12" overhang) Rise to hub: 20" Total rafter length: 76" (6'-4")	RING BEAM MITER + CORBEL + FOOTING RING BEAM: 30° miter both ends Hex exterior angle = 60°; miter saw set to 30° Both ends same miter direction (parallel) Dry-fit all 6 beams before hardware installation CORBEL (4x4): Both ends: 45° plumb cut only (no bevel) Body length: 16.97" diagonal Cut list: 21" Bolts: (2) 1/2" x 5" carriage bolts each end FOOTING: Auger 18" dia x 30"+ deep. Sonotube 24". 25 MPa concrete. J-bolt set wet. ABA66Z base.
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HARDWARE SCHEDULE — ORDER WITH LUMBER

ITEM	SPEC	QTY	PURPOSE
Simpson ABA66Z	ZMAX galvanized adjustable standoff post base	6	Post-to-footing anchor
J-Bolt 1/2" x 10"	Galvanized J-bolt + nut + washer	6	Set in concrete caisson
Simpson BC6 Post Cap	Black powder coat, post cap for 6x6 post / 6x8 beam	6	Post-to-beam erection + uplift
Knife Plate — A36 Steel	3/8" thick, laser cut to template per joinery-spec.json	12	Post-to-beam moment connection
1/2" x 5" Galv Carriage Bolt	+ 2" sq plate washer + nut — ASTM A307 galvanized	48	Corbel-to-post and corbel-to-beam
1/2" x 5" Galv Struct. Bolt	+ 2" sq plate washer + nut — galvanized hex head	24	Knife plate-to-post and knife plate-to-beam
SDS 1/2" x 6" Structural Screw	Simpson SDS25600 or equal hex-head structural lag	12	Rafter-to-hub (2 per rafter)
Simpson H2.5A Hurricane Tie	Galvanized rafter tie, 10d nails	6	Rafter-to-beam uplift (1 per rafter)

GENERAL NOTES FOR SHOP AND SITE

- Mark all lumber crown-side before cutting. Orient crowns up on rafters and beams.
 - Pre-drill all bolt holes. Avoid splitting ends: maintain 1-3/4" minimum edge distance on all bolts.
 - All exterior hardware to be hot-dip galvanized (ASTM A153) or stainless. No electro-galv in direct contact with cedar.
 - Apply oil-based end-grain sealer (Anchorseal or equal) to all fresh cuts within 24 hours of cutting.
 - Verify all 6 footing locations with laser level and string lines before pouring concrete. Check all diagonals equal (8'-8" approx).
 - Install hub as last structural element. Dry-fit all 6 rafters before driving SDS screws. Adjust hub height on temporary post if needed.
 - This is a no-permit structure per Saanich BCBC 2018 exemption threshold. Builder verifies final jurisdiction requirements before starting.
- Drawing set ref: D01 Plan | D02 Face Elev | D03 Corner Elev | D04 Rafter+Hub | D05 Joinery | D06 Footing | SB01 Cut List